Problem-Solution Fit

Date	12 August, 2026
Team ID	Individual
Project Name	AI Knowledge Retention Predictor
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] University students, competitive exam aspirants (e.g., GATE, GRE, UPSC), and self-paced online learners managing large volumes of technical coursework.	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** Limited study time, zero budget for expensive subscription tools, varying internet connection quality, and reliance on mobile/laptop web browsers.	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** **Anki / Flashcards:** *Pros:* Effective spaced repetition. *Cons:* Manual card creation is tedious and lacks interactive AI explanations. **Standard Quiz Apps:** *Pros:* Ready-made questions. *Cons:* Static, non-adaptive, and ignores individual forgetting curves.
2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]** University students, competitive exam aspirants (e.g., GATE, GRE, UPSC), and self-paced online learners managing large	**9. PROBLEM ROOT / CAUSE [RC]** Human cognitive reliance on the Ebbinghaus forgetting curve without structured, personalized revision schedules to reinforce	**7. BEHAVIOR** + ITS INTENSITY **[BE]** Binge-studying material 24–48 hours before assessments rather than maintaining consistent revision intervals. Intensity:

<table>
<tr><td>volumes of technical coursework.</td><td>long-term memory consolidation.</td><td>High stress and burn-out during exam weeks</td></tr>
<tr><td>3. TRIGGERS TO ACT [TR]

Low scores on practice tests, upcoming midterms or semester finals, and anxiety from forgetting core concepts previously mastered.</td><td rowspan="2">10. YOUR SOLUTION [SL]

AI Knowledge Retention Predictor: An interactive Streamlit application that calculates mathematical forgetting curves, dynamically generates tailored quizzes via Google Gemini API, and predicts optimal revision dates to maximize long-term retention.</td><td>8. CHANNELS OF BEHAVIOR [CH]

Online: Web search, YouTube tutorials, Discord/Telegram study groups, and Streamlit web applications.</td></tr>
<tr><td>4. EMOTIONS BEFORE / AFTER [EM]

Before: Overwhelmed, anxious, frustrated by repetitive memory loss, and uncertain about study readiness.

After: Confident, organized, reassured by visual mastery metrics, and clear on revision priorities.</td><td>Offline: Handwritten notes, physical textbooks, and peer discussion groups.</td></tr>
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